

- 7 (a) (i) the direction of the fields is the same OR fields are uniform OR constant electric field strength OR $E = V / d$ with symbols explained B1 [1]
- (ii) reduce p.d. across plates B1
increase separation of plates B1 [2]
- (iii) α opposite charge to β (as deflection in opposite direction) B1
 β has a range of velocities OR energies (as different deflections) and
 α all have same velocity OR energy (as constant deflection) B1
 α are more massive (as deflection is less for greater field strength) B1 [3]
- (b) $W = 234$ and $X = 90$ B1
 $Y = 4$ and $Z = 2$ B1 [2]
- (c) $A = 32$ and $B = 16$ and $C = 0$ and $D = -1$ B1 [1]